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AWS Project :

ScaleX – Scalable AWS web infrastructure with auto scaling

Step 1 : Create EC2 Instance : (webserver-1)

Instances (12) Info

Last updated 11 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐		i-0ad42454948f79ae5	Terminated	t3.micro	–	View alarms +	ap-south-1b	–	–	–
☐		i-095ddcc8319b5a944	Terminated	t3.micro	–	View alarms +	ap-south-1b	–	–	–
☐		i-06505418c2a337f25	Running	t3.micro	Initializing	View alarms +	ap-south-1b	ec2-13-203-201-191.ap...	13.203.201.191	–
☐	sns-ec2-instance	i-06415225816a83ed5	Running	t3.micro	3/3 checks pass	View alarms +	ap-south-1a	ec2-3-6-41-161.ap-sout...	3.6.41.161	–
☐	WebServer-1	i-085ca597d082f2975	Running	t3.micro	3/3 checks pass	View alarms +	ap-south-1a	ec2-3-108-223-136.ap-...	3.108.223.136	–
☐		i-078c0d9cf1ad9c3f1	Terminated	t3.micro	–	View alarms +	ap-south-1a	–	–	–
☐		i-0d3e9efd04cdfb9f	Terminated	t3.micro	–	View alarms +	ap-south-1a	–	–	–

Step 2 : Create AMI

Amazon Machine Images (AMIs) (1/1) Info

Owned by me

Find AMI by attribute or tag

Recycle Bin

EC2 Image Builder

Actions

Launch instance from AMI

☒	Name	AMI name	AMI ID	Source	Owner	Visibility	Status	Creation date
☒		WebServer-AMI	ami-08ce9e89f6aeb791c	216153283160/WebServer-AMI	216153283160	Private	Available	2026/04/22 21:57 GMT+

AMI summary for ami-08ce9e89f6aeb791c

AMI ID: ami-08ce9e89f6aeb791c
AMI name: WebServer-AMI
Root device name: /dev/xvda
Boot mode: uefi-preferred
Description: –
Last launched time: –
Source AMI ID: ami-0e12ffc2dd465f6e4

Image type: machine
Owner account ID: 216153283160
Status: Available
State reason: –
Product codes: –
Block devices: /dev/xvda: snap-0ea4ed394ad2fde0: 8:true:gp3
Source AMI Region: ap-south-1

Platform details: Linux/UNIX
Architecture: x86_64
Source: 216153283160/WebServer-AMI
Creation date: 2026-04-22T16:27:03.000Z
RAM disk ID: –
Deregistration protection: Disabled

Root device type: EBS
Usage operation: RunInstances
Virtualization type: hvm
Kernel ID: –
Deprecation time: –
Allowed image: –

Permissions: Storage My AMI usage AMI ancestry Tags

Image share permission: Private
This image is only shared with account IDs, organizations, or OUs that you have specified.

Restrictions for sharing images publicly are managed using Block public access for AMIs setting under Data protection and security.

Shared accounts

Step 3 : Connect with SSH and enter a commands :

```
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.2.x86_64      apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch      httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-34-117 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-34-117 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-34-117 ~]$ echo "Hello Vivek" > /var/www/html/index.html
Hello Vivek
[ec2-user@ip-172-31-34-117 ~]$ curl http://ec2-3-108-223-136.ap-south-1.compute.amazonaws.com
Connection to ec2-3-108-223-136.ap-south-1.compute.amazonaws.com closed.
PS C:\Users\Vivek Chaudhari\OneDrive\Desktop\aws-project>
```

Step 4 : Create Target Group

my-target-group

my-target-group Actions

Details
am:aws:elasticloadbalancing:ap-south-1:216153283160:targetgroup/my-target-group/e9a1237c20b0bab3

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-07abdf21c0b27f179
IP address type IPv4	Load balancer None associated		

1	1 Healthy	0 Unhealthy	0 Unused	0 Initial	0 Draining
Total targets					

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

Step 5 : Create Load Balancer

Load balancers > My-ALB

My-ALB Actions

Details

Load balancer type Application	Status Active	VPC vpc-07abdf21c0b27f179	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZP978AFLXTNZK	Availability Zones subnet-09a155f615754ca22 ap-south-1b (apps1-ac3) subnet-05bab617631c7f6b9 ap-south-1a (apps1-ac1)	Date created April 22, 2026, 22:07 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:216153283160:loadbalancer/app/My-ALB/fc93c1a3b48acde5		DNS name info My-ALB-1147062723.ap-south-1.elb.amazonaws.com (A Record)	

Listeners and rules (1) Manage rules Manage listener Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
☐ HTTP:80	Forward to target group my-target-group 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable

Step 6 : Create Launch Template

WebTemplate

WebTemplate (lt-0ebc693df00b74f4d)

ActionsDelete template

Launch template details

Launch template ID

lt-0ebc693df00b74f4d

Launch template name

WebTemplate

Default version

1

Owner

arn:aws:iam::216153283160:root

DetailsVersionsTemplate tags

Launch template version details

ActionsDelete template version

Version

1 (Default)

Description

-

Date created

2026-04-22T16:59:37.000Z

Created by

arn:aws:iam::216153283160:root

Instance detailsStorageResource tagsNetwork interfacesAdvanced details

AMI ID

ami-08ce9e89f6aeb791c

Instance type

t3.micro

Availability Zone

-

Availability Zone Id

-

Key pair name

WebServer-key

Security groups

-

Security group IDs

sg-0c65a4510a9c43834

Step 7 : Create Auto Scaling (SNS Enable for Email Notification)

Auto Scaling groups > web-template-autoscalineg

web-template-autoscalineg

web-template-autoscalineg Capacity overview

Edit

arn:aws:autoscaling:ap-south-1:216153283160:autoScalingGroup:ee09e817-e220-4a02-b659-7d48b3095a12:autoScalingGroupName/web-template-autoscalineg

Desired capacity

2

Scaling limits

1 - 2

Desired capacity type

Units (number of instances)

Status

Updating capacity

Date created

Wed Apr 22 2026 22:43:22 GMT+0530 (India Standard Time)

Final Output :

← → ↺ ⚠ Not secure http://my-alb-1147062723.ap-south-1.elb.amazonaws.com

Hello Vivek 🚀 from EC2 Server

This project successfully implements a scalable and highly available web application architecture on AWS using EC2, Application Load Balancer, and Auto Scaling. The system distributes incoming traffic efficiently across multiple instances, automatically adjusts resources based on demand, and ensures minimal downtime through health checks and failover mechanisms. Additionally, SNS integration provides real-time notifications for instance launch and termination events, making the system reliable, automated, and production-ready.

SNS Notification :



AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

23:06 (27 minutes ago) ☆ 😊 ↶ ⋮

Service: AWS Auto Scaling
Time: 2026-04-22T17:36:31.380Z
RequestId: acf676d8-fce0-52b2-1363-701ddd56f39a
Event: autoscaling:EC2_INSTANCE_TERMINATE
AccountId: 216153283160
AutoScalingGroupName: web-template-autoscalineg
AutoScalingGroupARN: arn:aws:autoscaling:ap-south-1:216153283160:autoScalingGroup:ee09e817-e220-4a02-b659-7d48b3095a12:autoScalingGroupName/web-template-autoscalineg
ActivityId: acf676d8-fce0-52b2-1363-701ddd56f39a
Description: Terminating EC2 instance: i-095ddcc8319b5a944
Cause: At 2026-04-22T17:31:29Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.
StartTime: 2026-04-22T17:31:29.684Z
EndTime: 2026-04-22T17:36:31.380Z
StatusCode:InProgress
StatusMessage:
Progress: 50
EC2InstanceId: i-095ddcc8319b5a944

...